# CS3003: IT Project Management

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# CS3003: IT Project Management

## Part 4: The first/basic Project Planning Elements:

Scope *(discussed),*

Time *(discussed)*

**Resources,**

Communications

# Resources Planning

- Plan for the resources that are required to successfully carry out the project
- Main **Types of Resources**:
  - **People**, with specific Skills -> who will do what?
  - **Equipment** (incl. all elements of the HW infrastructure),
  - **Software Infrastructure and Tools**,
  - **Facilities** (e.g. Space)
- Primary Inputs:
  - The activity list and corresponding resource requirements (e.g. skills, infrastructure, tools, equipment)
  - The Project Stakeholders Analysis and Governance Structure and Roles
  - Resources' Availability
  - Trade-off between tasks Time and Resources (an iterative process)
  - Personalities, Experience, Group Dynamics, People Growth
  - Related Risks

Key Question: How will I get the best resources and use who I have as best as I can?

# Resources Planning: Example Resource Requirements Analysis

| **Activity** | **Skills/Knowledge required** (including number of) | **Potential Personnel** (possibly including availability) | **Other Resources** (e.g. equipment) |
|---|---|---|---|
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# Resources Planning: Some Tools and Techniques

- Responsibility Matrix
- Resource Gantt Chart
- Resource Loading Histogram
- Resource Loading Table and Resource Leveling

# Resources Planning: Some Tools and Techniques

- **Responsibility Matrix** or RACI matrix

| ACTIVITY \ RESOURCE | Joe PM | Bill TL | Jill Inf.Arch. | Kyle Progr. | Claire GA | Ron Q/A | Liam Client | Trin Client |
|---|---|---|---|---|---|---|---|---|
| Draft software blueprint | | A | R | C | C | C | I | I |
| Meet with client to refine blueprint | A | R | C | I | I | | C | C |
| Refine software architecture diagram | | A | R | C | C | C | I | I |
| Design navigation structure | | A | R | C | C | C | I | I |
| Design graphical interface | I | A | C | C | R | I | I | I |
| Program prototype | I | C | A | R | I | C | | |
| Review prototype with client | R | C | C | I | I | I | R | R |

R=Responsible A=Accountable /Approved C=Consulted I=Informed

# Resources Planning: Some Tools and Techniques

- **Resource Gantt Chart**

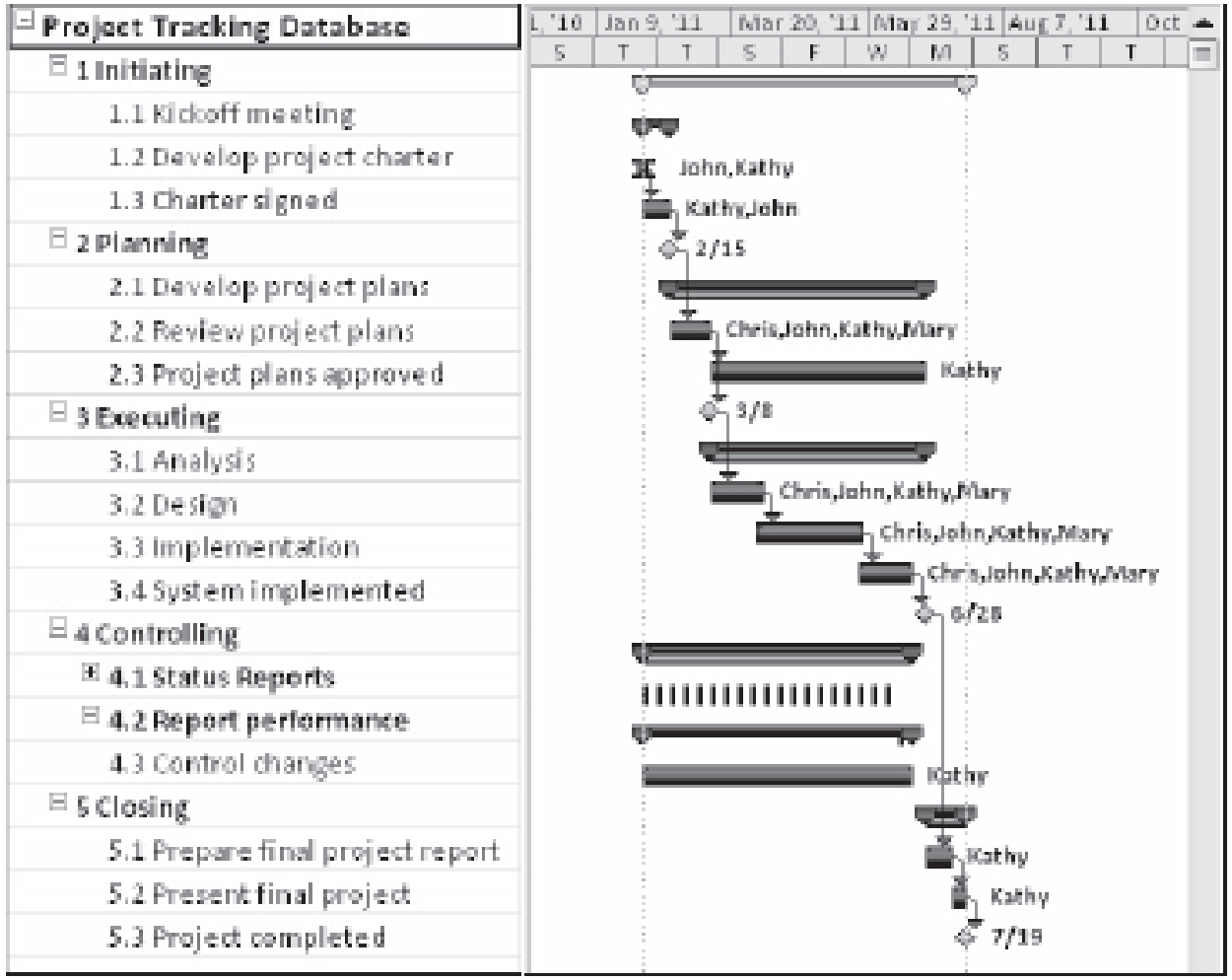


Image from Schwalbe's book .

# Resources Planning: Some Tools and Techniques

- **Resource Loading Histogram:** the amount of individual resources that an existing schedule requires over time

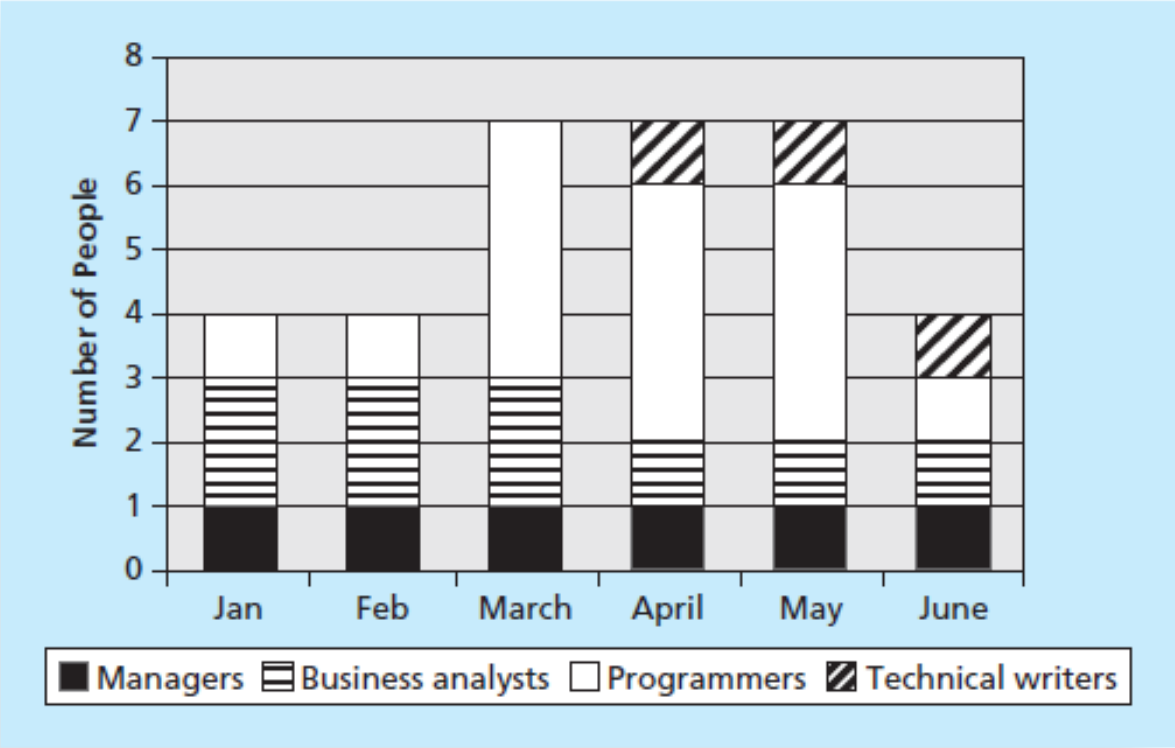


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FIGURE 9-6 Sample resource histogram

Image from Schwalbe's book .

# Resources Planning: Some Tools and Techniques

- **Resource Loading Table**
  - Item in red indicates resource overallocation
  - **Resource Leveling:** resolving resource conflicts (incl. overallocation) or achieving a smoother distribution of resources (mainly by delaying tasks based on their slack)

Tasks | Resources | Track | Report

| | | Resource Name | Work | Details | uary 1 | January 11 | | January 21 | February 1 | | February 11 | F |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | 1/4 | 1/11 | 1/18 | 1/25 | 2/1 | 2/8 | 2/15 | |
| 1 | | Tom | 40 hrs | Work | 32h | 8h | | | | | | |
| | | *Assign Bids* | *40 hrs* | Work | 32h | 8h | | | | | | |
| 2 | ! | **Jeff** | **80 hrs** | Work | | 64h | 16h | | | | | |
| | | *Document Awards* | *40 hrs* | Work | | 32h | 8h | | | | | |
| | | *Calculate Costs* | *40 hrs* | Work | | 32h | 8h | | | | | |
| 3 | | Sue | 8 hrs | Work | | | 8h | | | | | |
| | | *Select Winning Bid* | *8 hrs* | Work | | | 8h | | | | | |
| 4 | | Carol | 32 hrs | Work | | | 24h | 8h | | | | |
| | | *Develop PR Campaign* | *32 hrs* | Work | | | 24h | 8h | | | | |
| 5 | | Bob | 0 hrs | Work | | | | | | | | |
| | | | | Work | | | | | | | | |
| | | | | Work | | | | | | | | |

Image from Microsoft and Jeffrey Pinto's Book on Project Management .

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## Part 4: The first/basic Project Planning Elements:

Scope *(discussed),*

Time *(discussed)*

Resources (*discussed*)

**Communications**

# Communications Planning

- **Communications *Management***: The processes required to ensure timely and appropriate generation, collection, distribution, storage, retrieval and disposition of project information. Goal is that essential knowledge is passed to team members and stakeholders so that decisions are based on accurate and complete information.
- **Communications *Plan***: a document that describes project communications. Addresses the following items:
  - Stakeholders Communications Requirements
  - Information to be communicated, incl. format, content and level of detail
  - Who will receive the information and who will produce it
  - Methods (e.g. reports, meetings) or technologies to distribute the information
  - Frequency of Communication
  - Escalation Procedures for Resolving Issues
  - Revision procedures for updating the communications management plan
  - Project Archives Information
  - Lessons Learned (shared across the organization)
  - Glossary/Terminology
- Answers the questions: What (information), Where (it will come from), Who, When?

# Communications Plan Sample Template

| Stakeholder | What | Format (Method) | How often (When) | Person Responsible |
|---|---|---|---|---|
| Steering Committee | Monthly status report | Written report via e-mail | monthly | PM |
| Sponsor | Sponsor status report | Written report via e-mail/Meeting | every two weeks | PM |
| PM | Weekly Progress report | Meeting | weekly | Team Members |
| Change Authority Members | Change Requests | Meeting | Ad hoc/as needed | PM |
| … | | | | |
| | | | | |
| | | | | |

# Communications Planning: Typical Project Meetings

- **Kick-off Meeting** (discussed – at end of initial planning or at end of full planning)
- **Core Team Project Meetings:**
  - PM is responsible for determining frequency (e.g. weekly or ad-hoc) and preparing/organizing
  - Every member of the project's core team should participate (or representatives in larger projects)
  - Agenda: accomplishments since the last meeting, status vs work plan, outstanding items since the last meeting, issues and resolution planning, scope/changes, risks, conclusions and actions items.
  - Meeting minutes are documented and distributed to the whole team. Changes may have to be made to schedule/plans etc.
- **Project Status and/or Progress Reports and/or Meetings** (typically regularly scheduled)
  - **Status:** where the project stands at a particular point in time (e.g. schedule, costs, scope)
  - **Progress:** what has been accomplished during a certain period (e.g. a month), plans for the next period, issues, risks, projected/proposed actions, changes, *forecasting* (predicting future status/performance)

  Different reports/meetings of such type may take place: with Sponsor, with key-stakeholders (e.g. Business Units), Steering Committee. Meeting minutes are documented and distributed. Changes may have to be made to schedule/plans etc.
- **Ad Hoc Meetings**
- **Project Closure Meeting:** to formally closeout the project. Agenda: Post-implementation audit, lessons-learned, release of resources, formal acceptance, celebration.

➢ What are similar meetings in Agile Process Methodologies?

# Communications Plan: Important Notes/Clarifications

***Planning***

**Plan Communication**

***Implementing***

**Distribute Information**

Make information available to project stakeholders, as defined by the communications plan, and respond to unexpected requests for information.

***Controlling***

**Manage Communications**

Ensure that the communications needs of project stakeholders are satisfied. Focus:

- Increase understanding and cooperation
- Timely, accurate, unbiased information
- Resolve communications issues, if and when they arise